SIAM CHAMNAN KIT

DevOps series

Development consult

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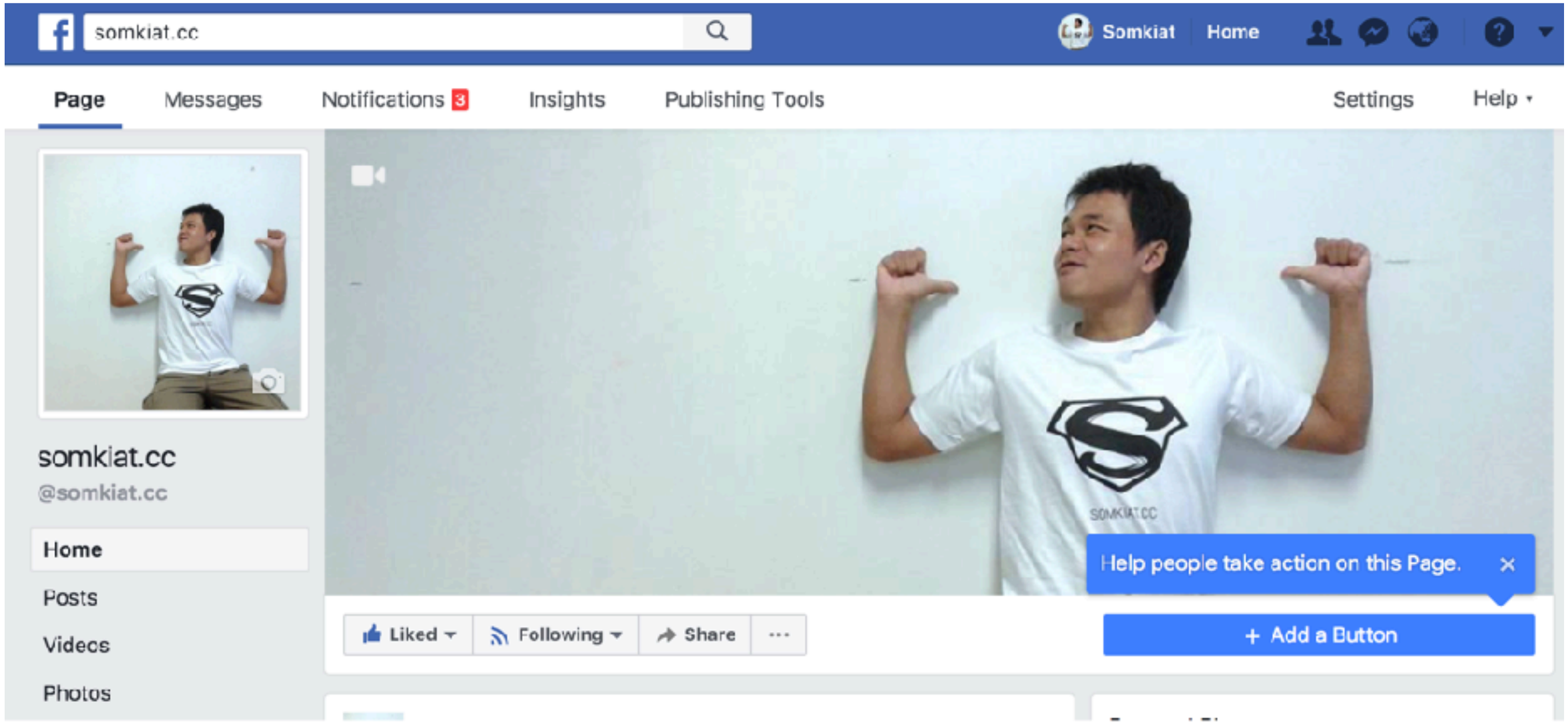

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Topics

1. Server environment setup
2. Dev -> UAT -> Production
3. Deployment strategies
4. Design your deployment pipeline
5. Develop your deployment pipeline
6. Workshop with docker

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https://github.com/up1/workshop-development-consult

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Server Environment Setup

Dev

UAT

Production

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https://12factor.net/

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https://12factor.net/

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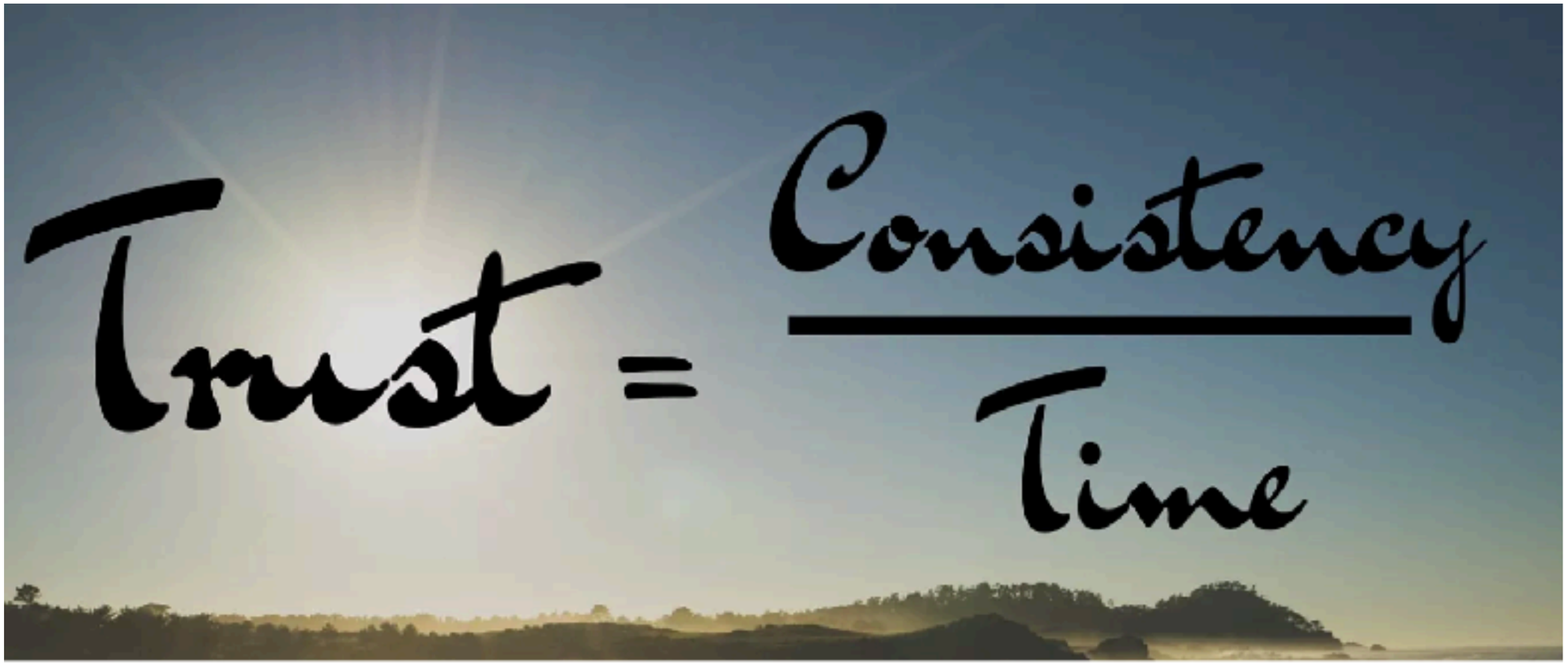

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Consistency Problems ?

Big problem of software failure

"It work on my machine"

Unpredictable Bugs	Deployment failures
Untestable scenarios	Security and compliance risks

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Why environment diff ?

Manual configuration
Cost constraints
Purpose mismatch

Dev for speed and experiment	**UAT** for validate business	**Production** for high availability

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Best Practices ?

IoC (Infrastructure as Code)

Containerization technology

CI/CD pipelines

Data masking for sensitive data

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DevSecOps Tools ?

https://digital.ai/learn/devsecops-periodic-table/

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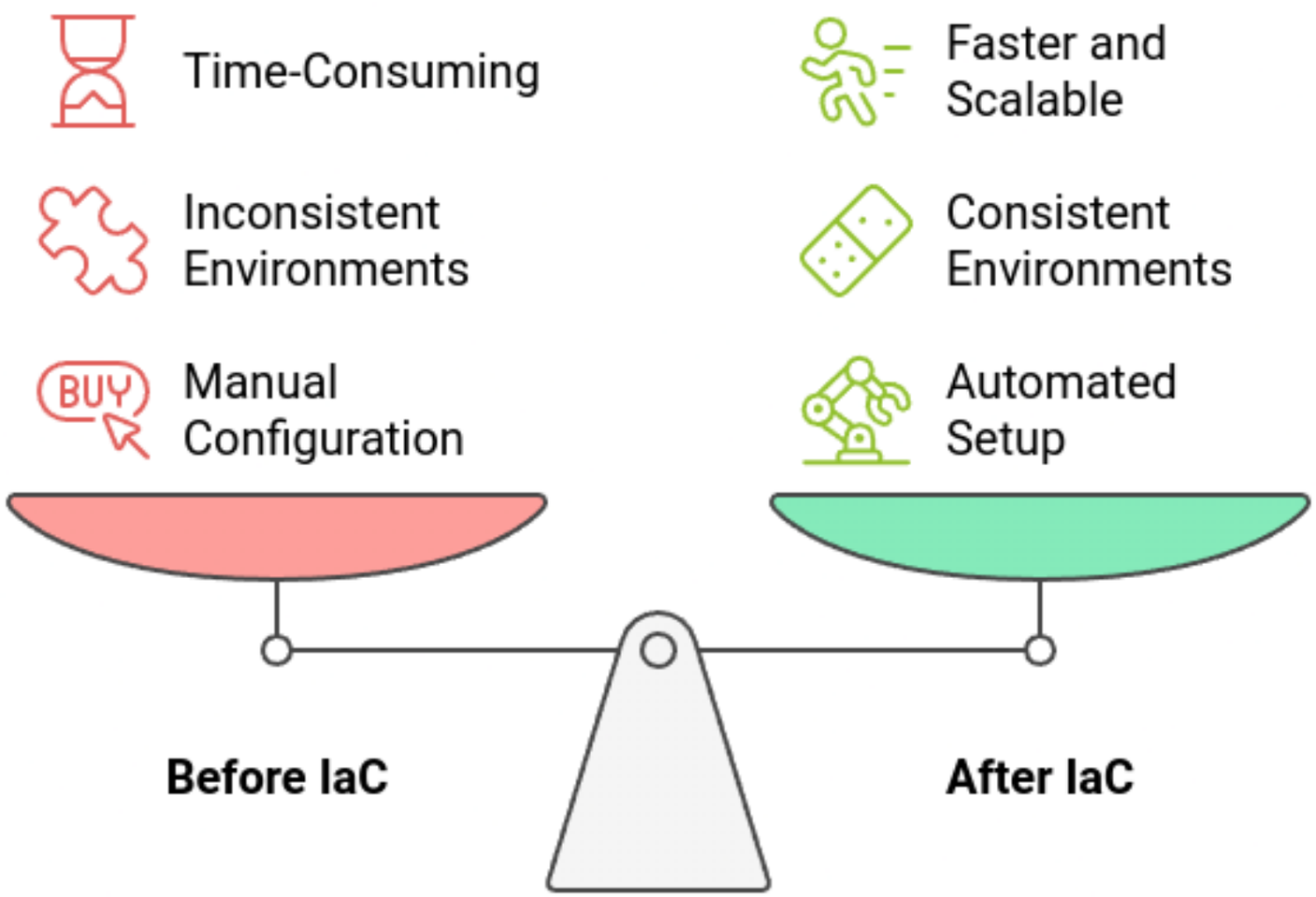
IoC ?



IaC transforms infrastructure management from manual to automated.

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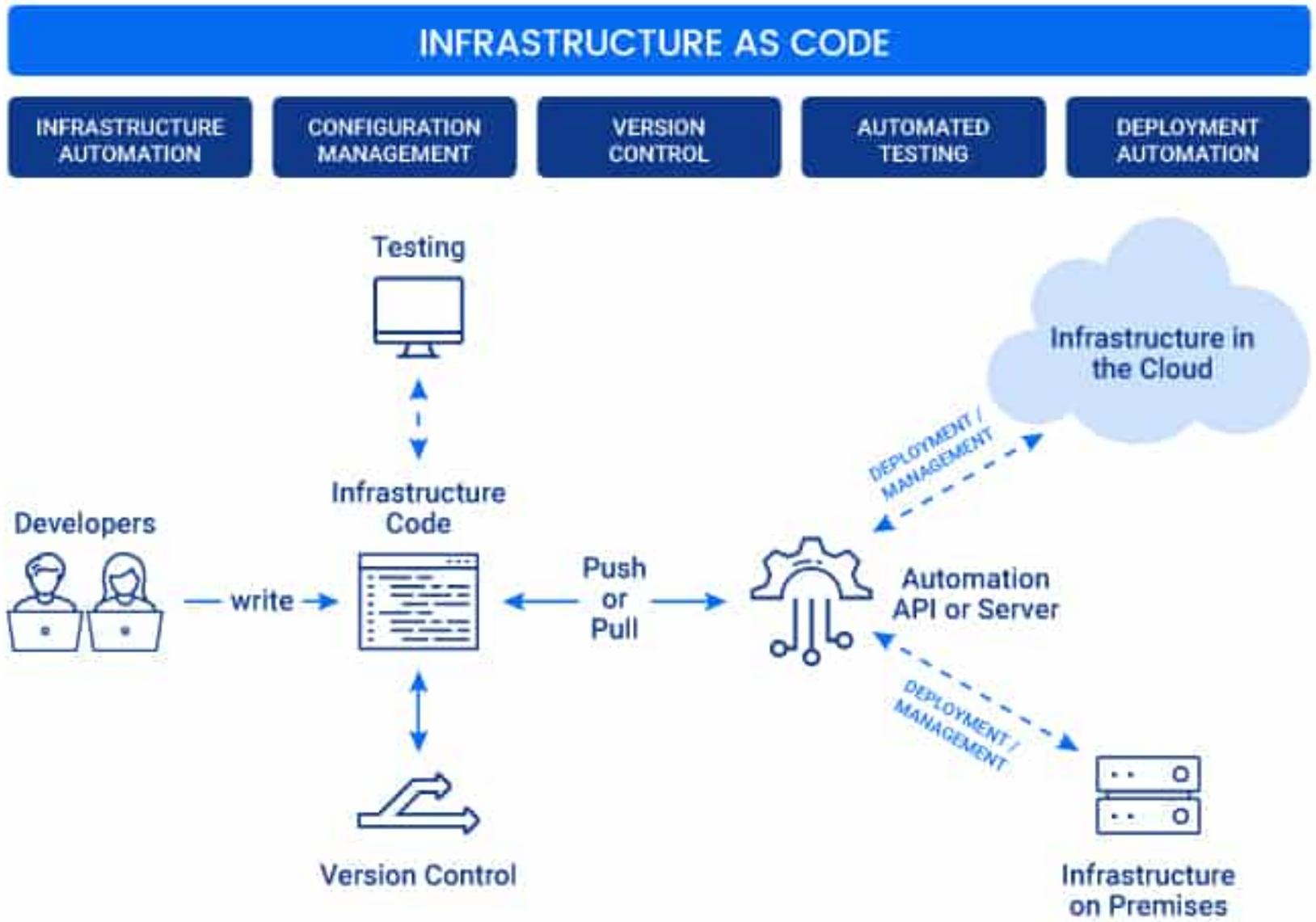
IoC works ?



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IoC Tools ?

ANSIBLE

HashiCorp Terraform

Pulumi

CHEF

perforce Puppet

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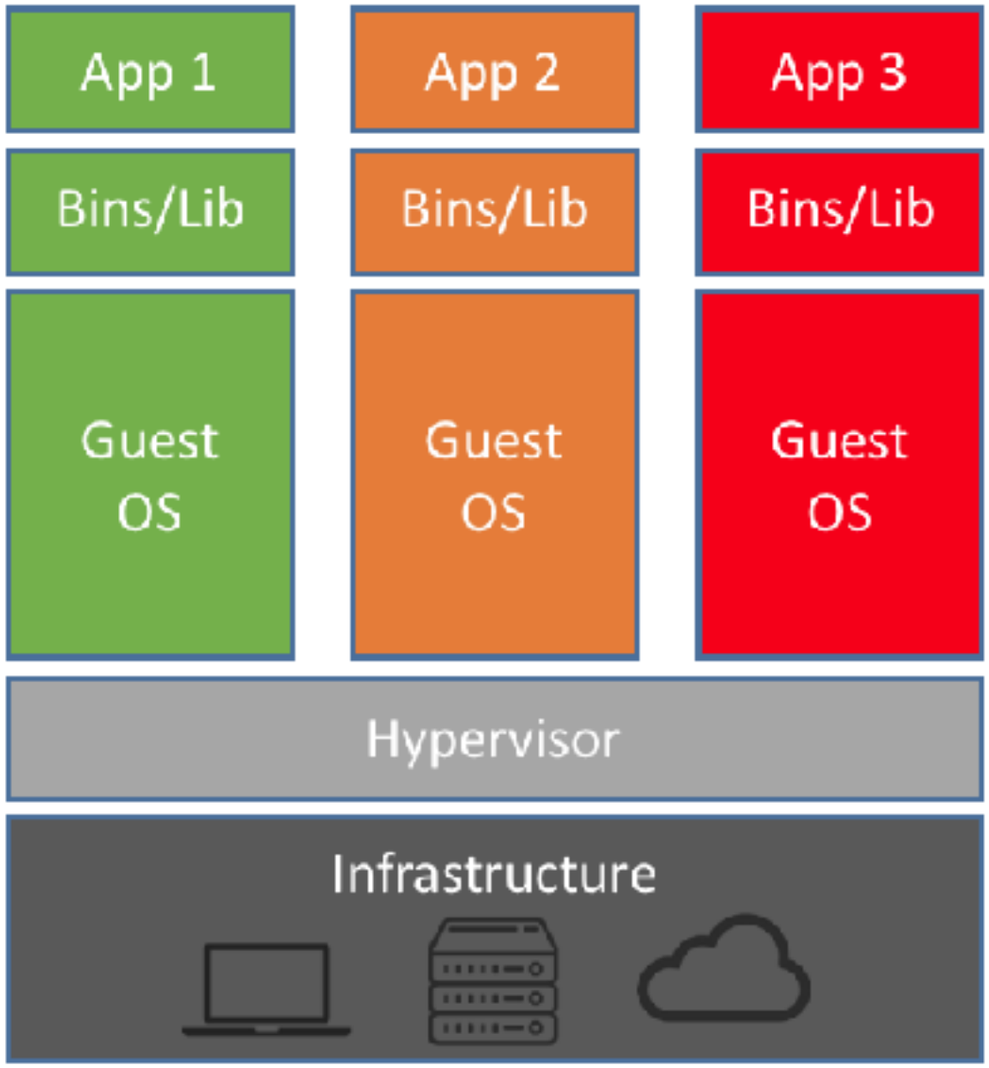
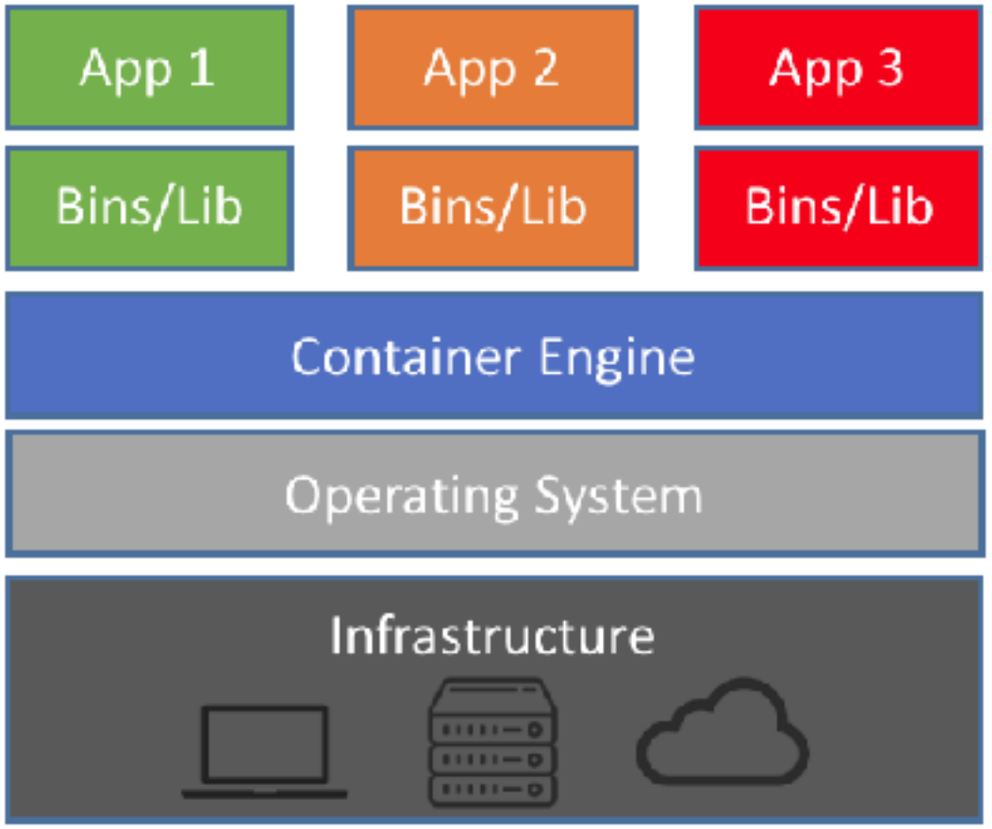
Containerization Technology ?



Machine Virtualization



Containers

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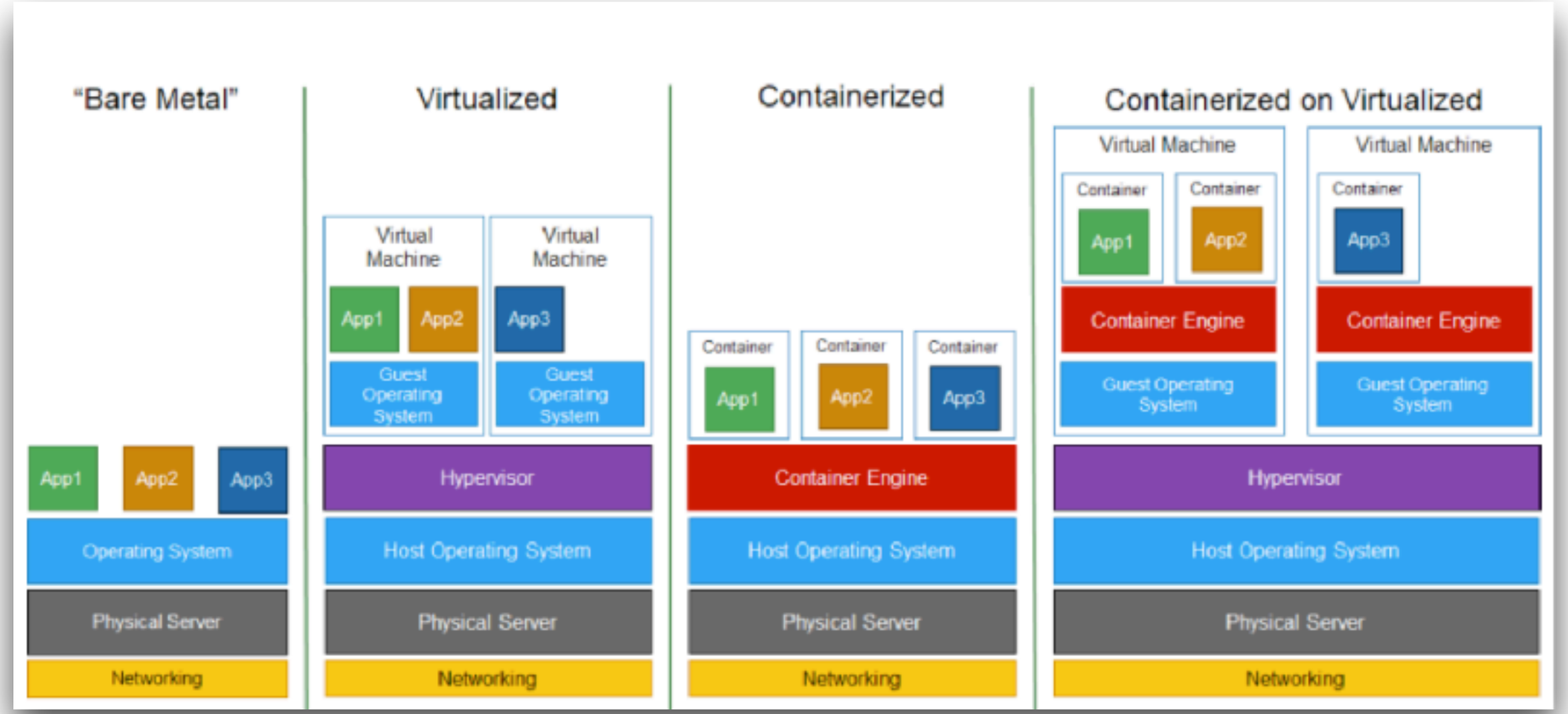
Containerization Technology ?



https://blog.bytebytego.com/p/what-are-the-differences-between

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Deployment Strategies

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Start with
Continuous Integration (CI)
Continuous Delivery (CD)

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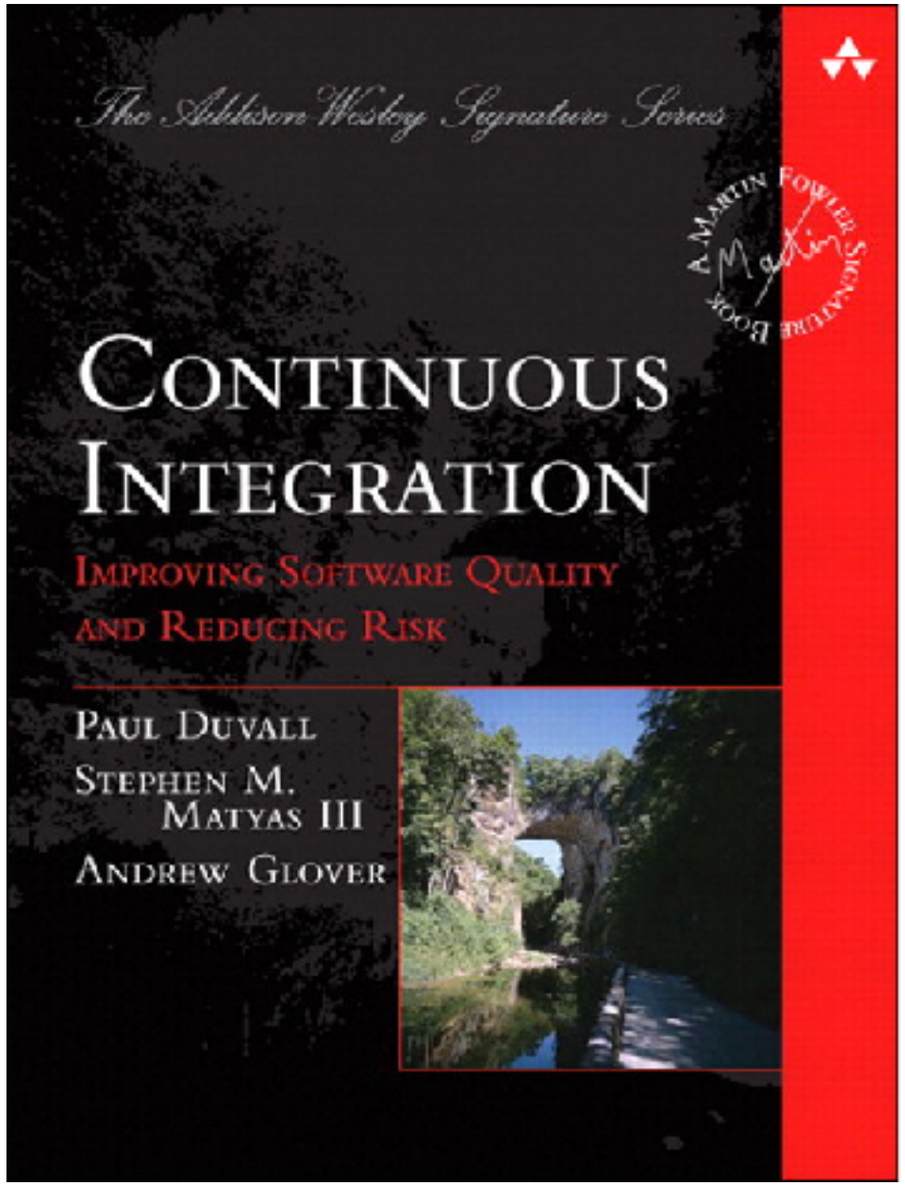
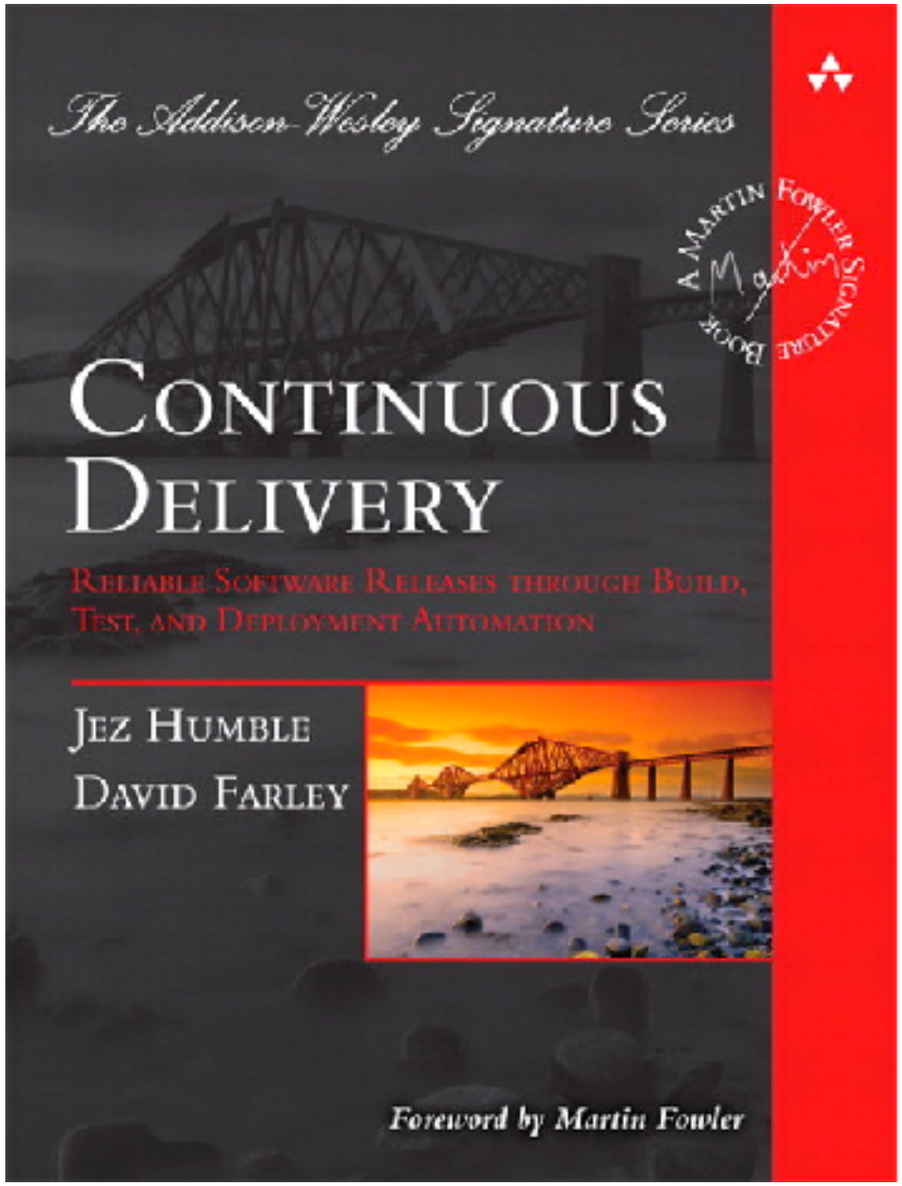
Improve quality and reduce risk





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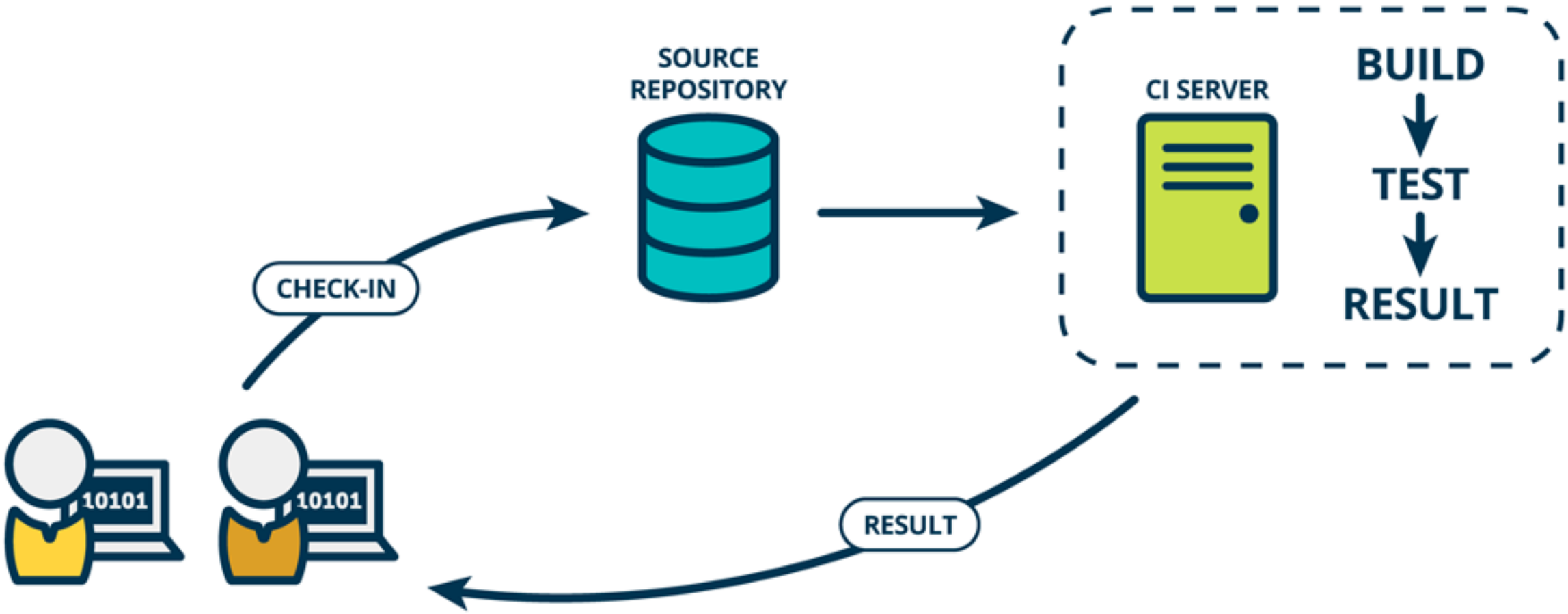
Continuous Integration



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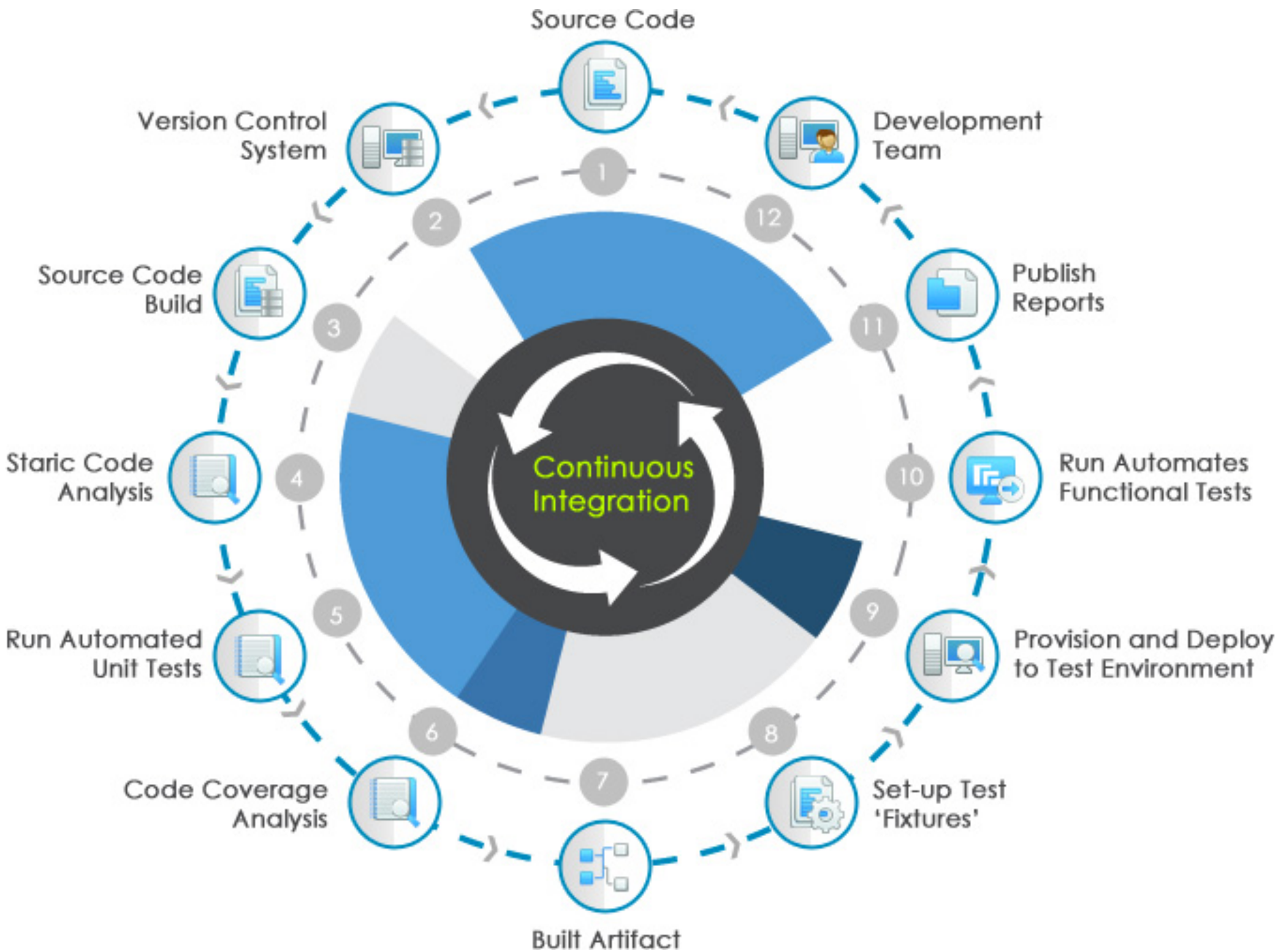

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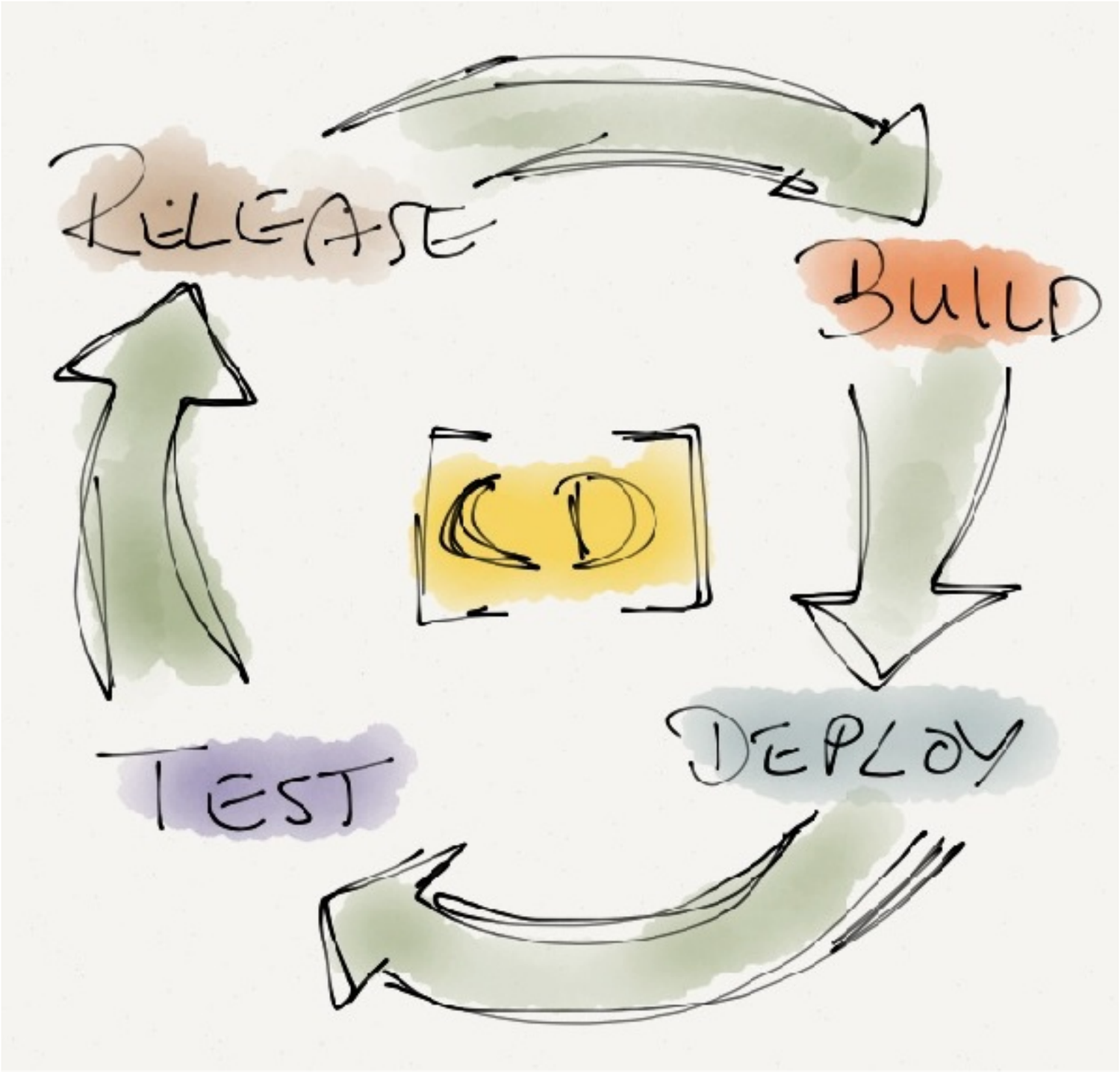
CD ?



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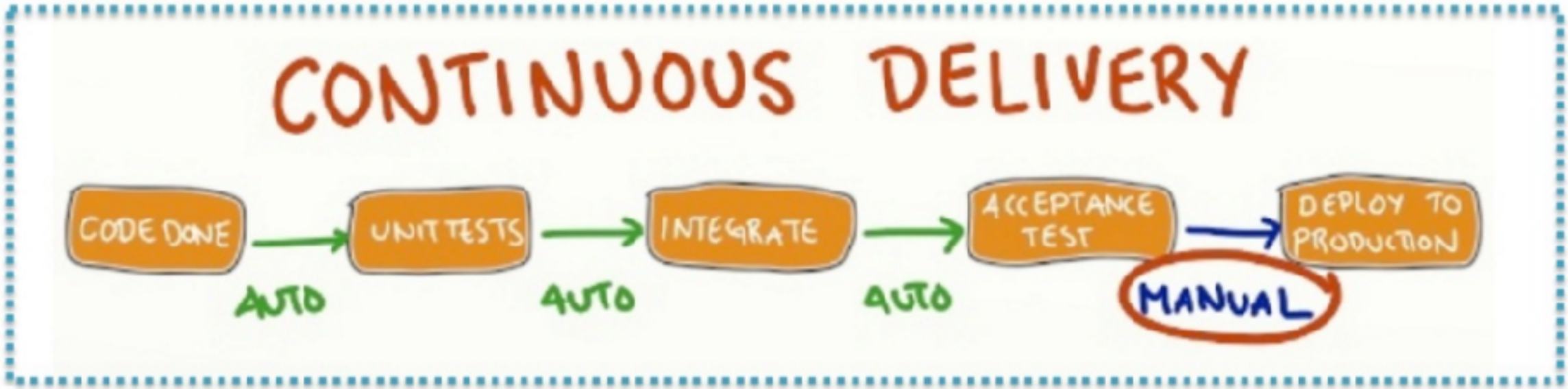
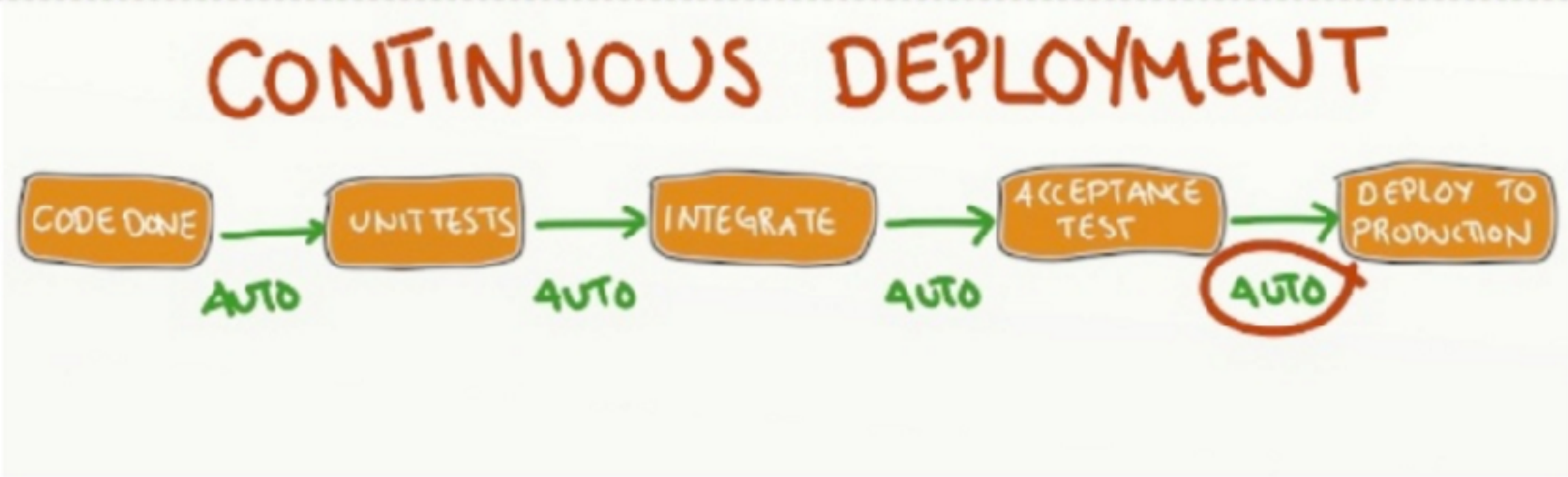
CD ?





http://blog.crisp.se/2013/02/05/yassalsundman/continuous-delivery-vs-continuous-deployment

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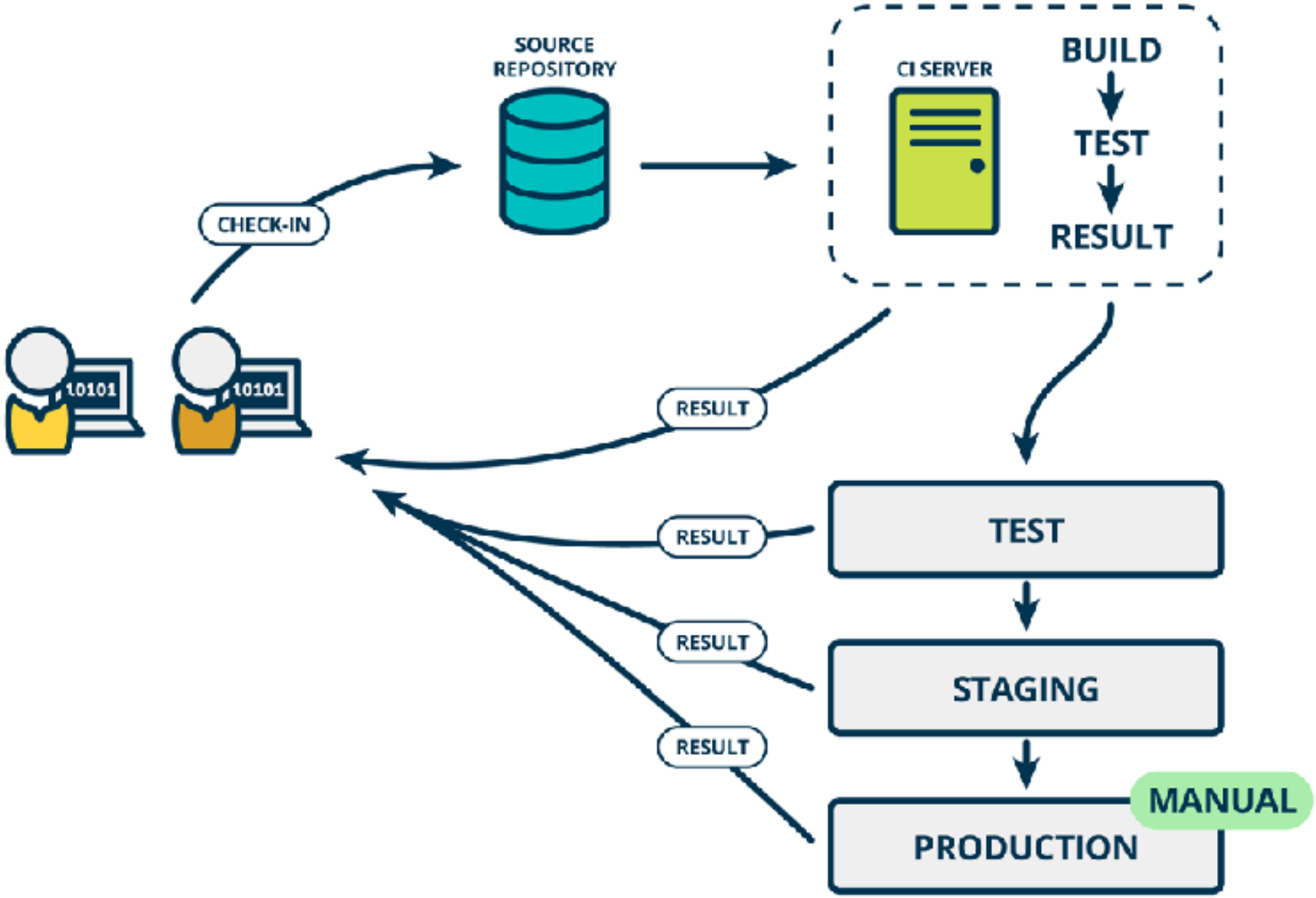
Continuous Delivery



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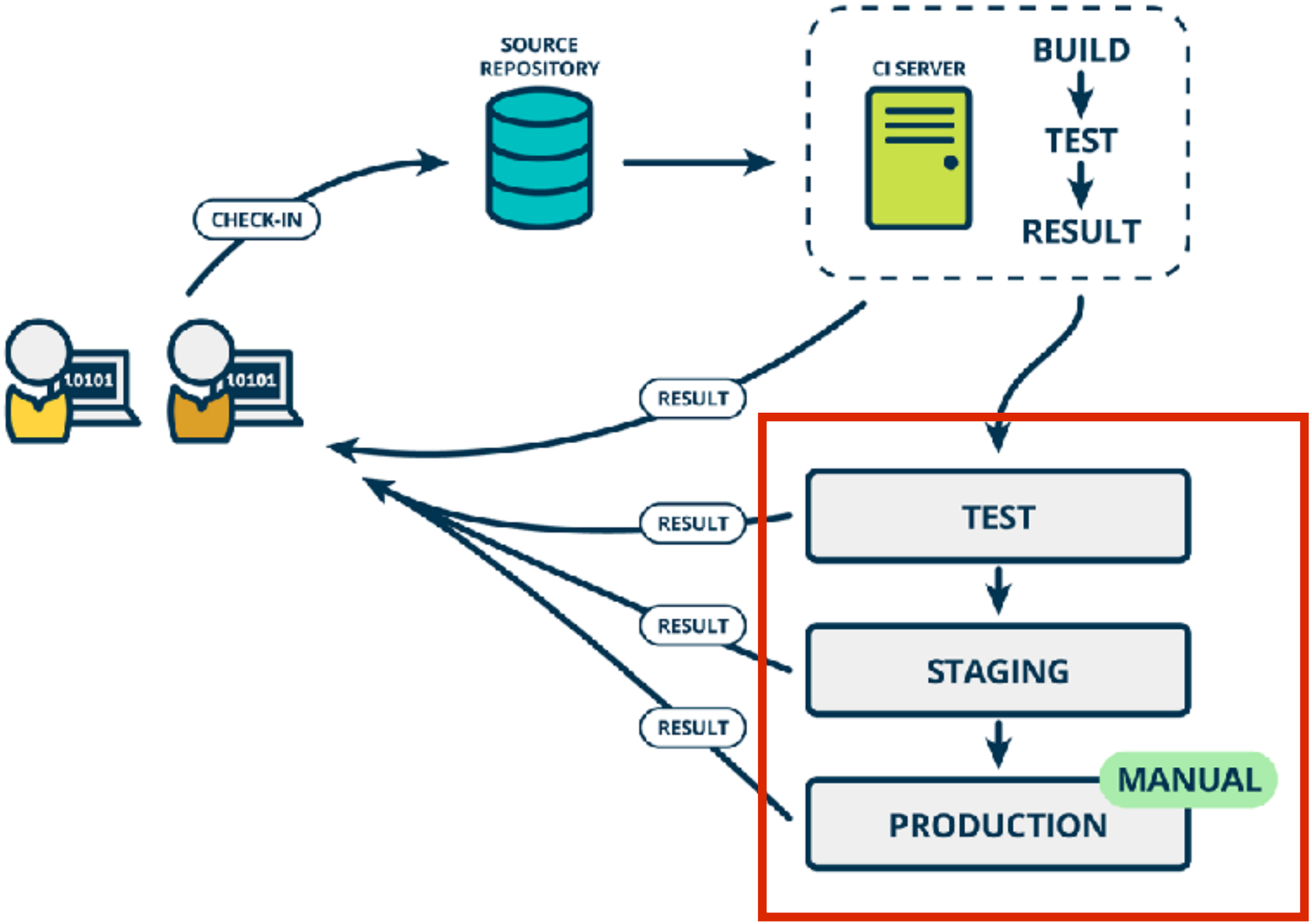
Rise of DevOps



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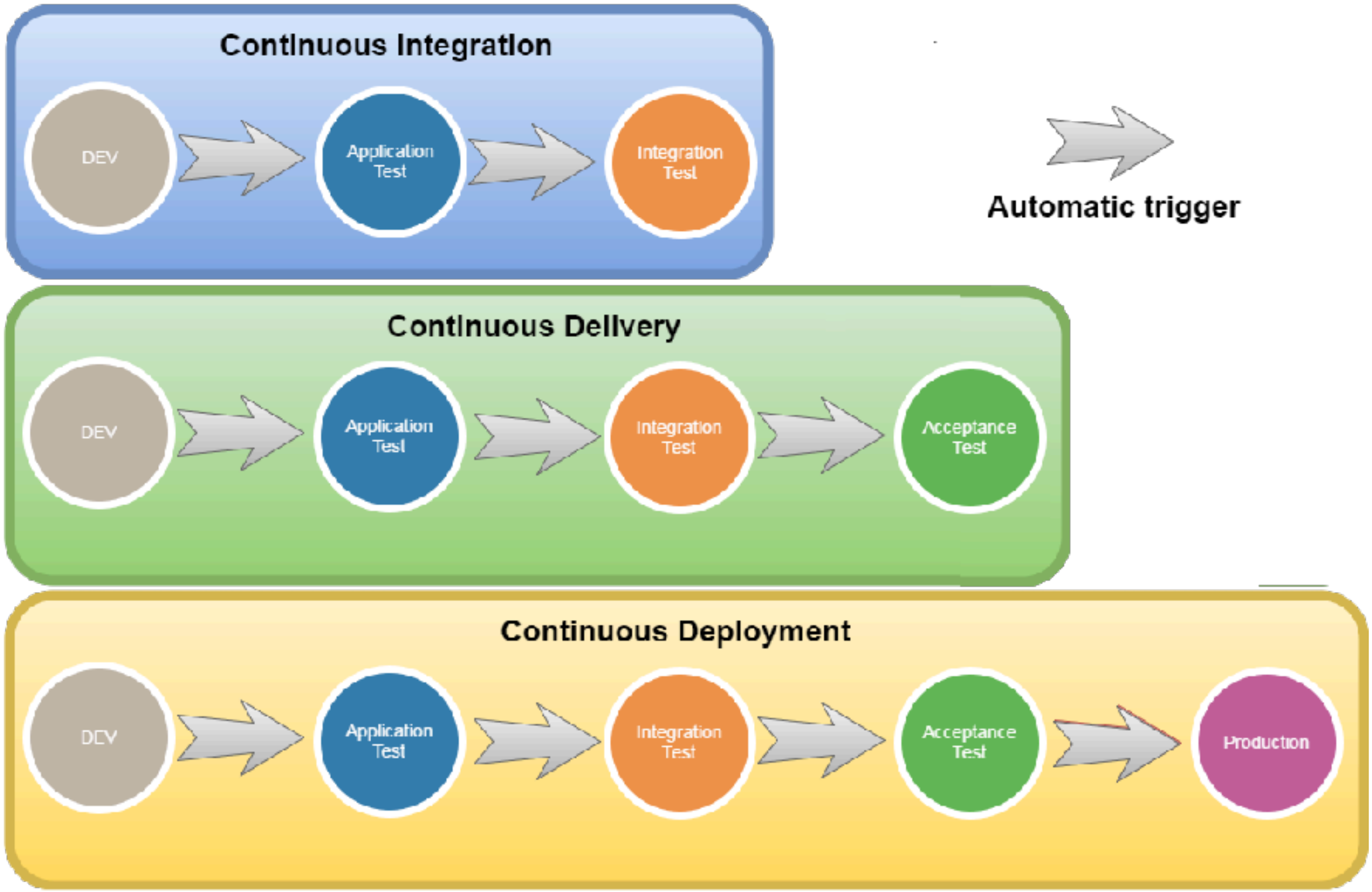

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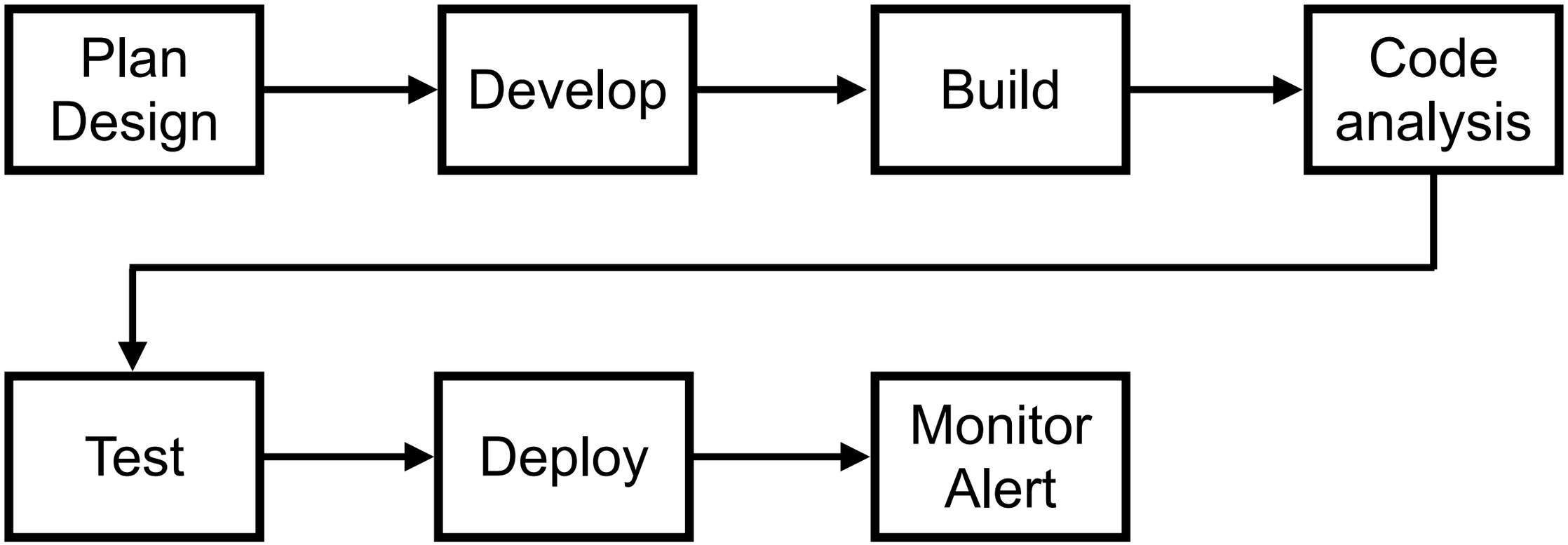
Design your Pipeline ?



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Deployment Strategies ?

Recreate (big bang)
Blue/Green deployment
Canary deployment
Feature toggle/flag

https://docs.aws.amazon.com/whitepapers/latest/introduction-devops-aws/deployment-strategies.html

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Recreate

The current is completely shutdown before the new is started

Simple to implementation

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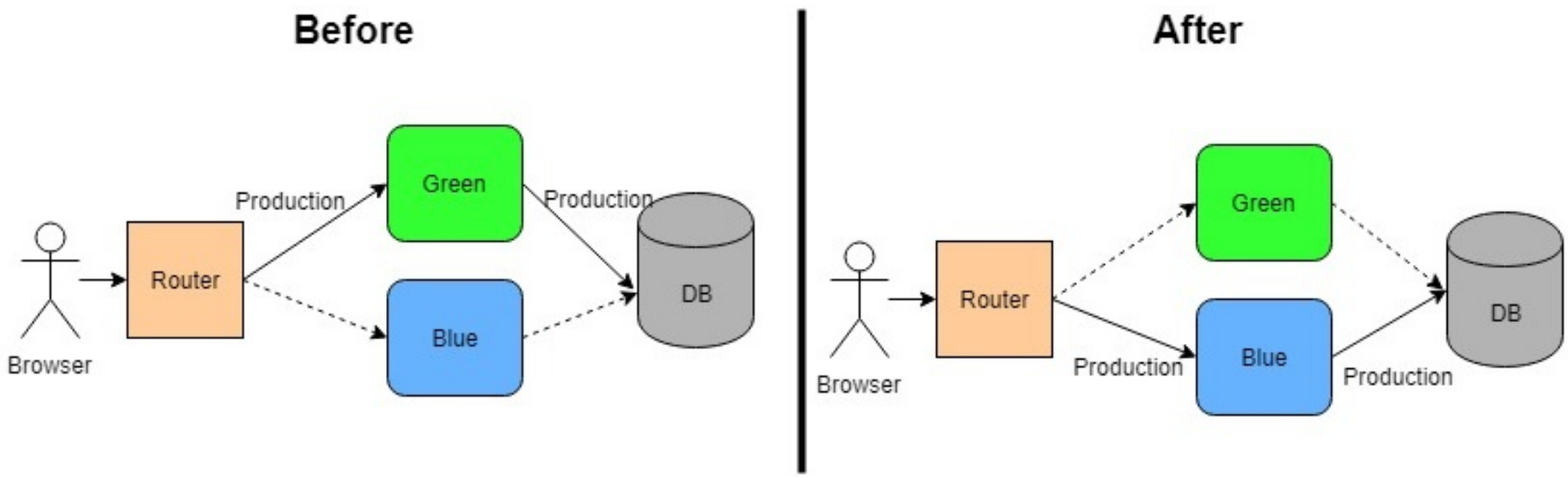
Blue/Green

Maintain 2 identical production environments
Blue = current/live
Green = new version



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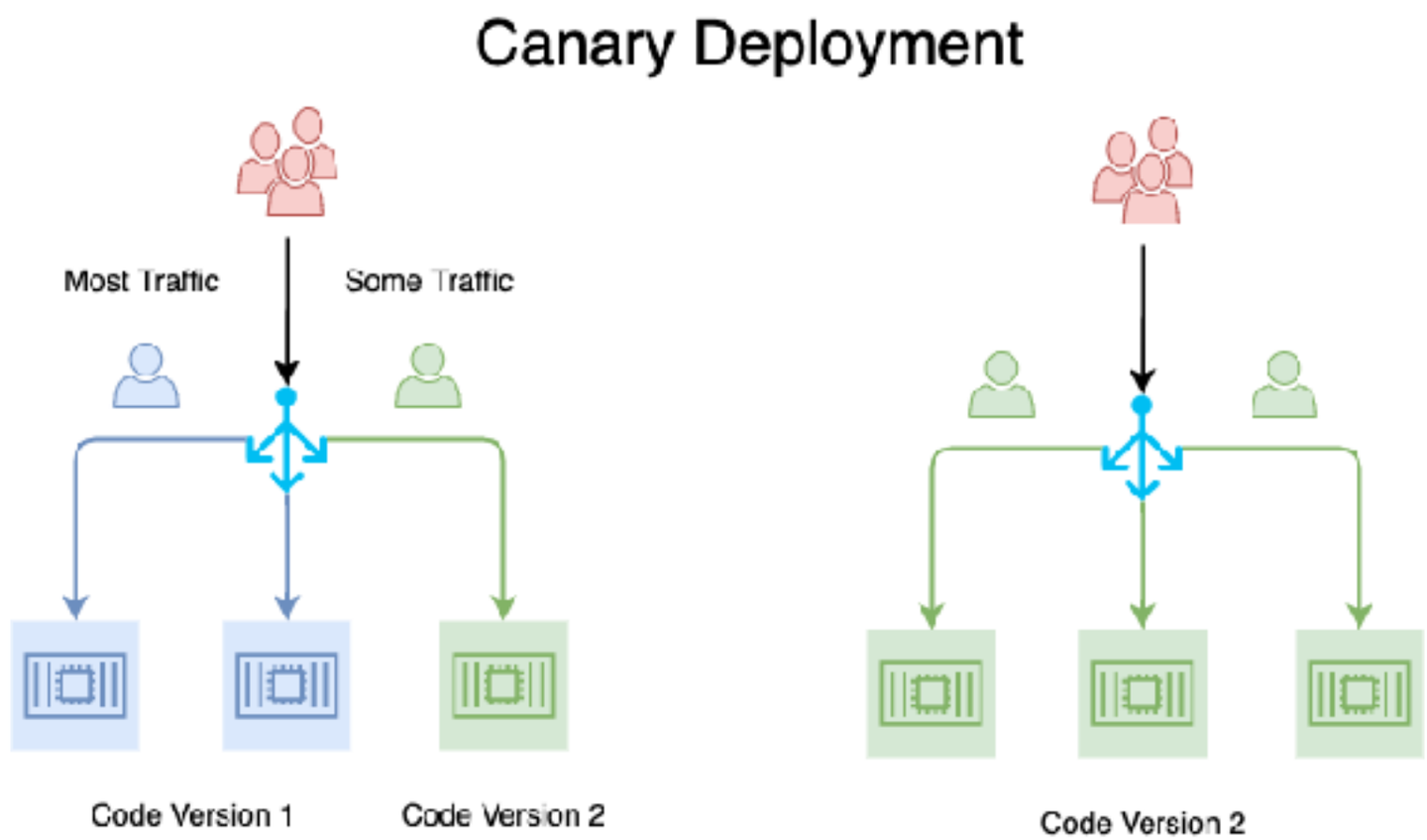
Canary

Like blue/green deployment
When rollout to new version, try to rollout in small group



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How to choose Strategies ?

No downtime need
High risk mitigation
Simple/internal app
Limited resources (cost)

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Deployment Strategies ?

Strategy	Downtime	Risk	Rollback speed	Cost
Recreate	High	High	Slow	Low
Blue/Green	None	Low	Instant	High
Canary	None	Very low	Fast	Medium

https://docs.aws.amazon.com/whitepapers/latest/introduction-devops-aws/deployment-strategies.html

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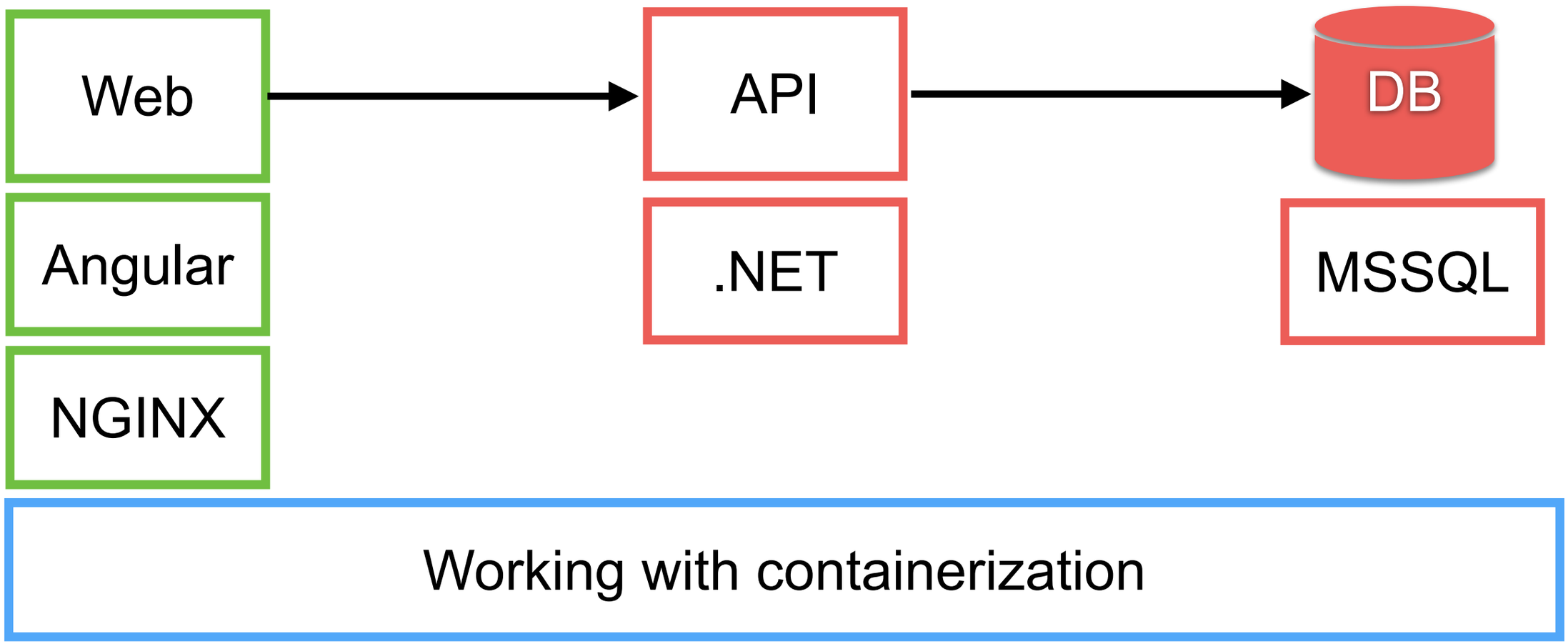
Simple App Architecture



https://github.com/up1/workshop-devops-angular-dotnet

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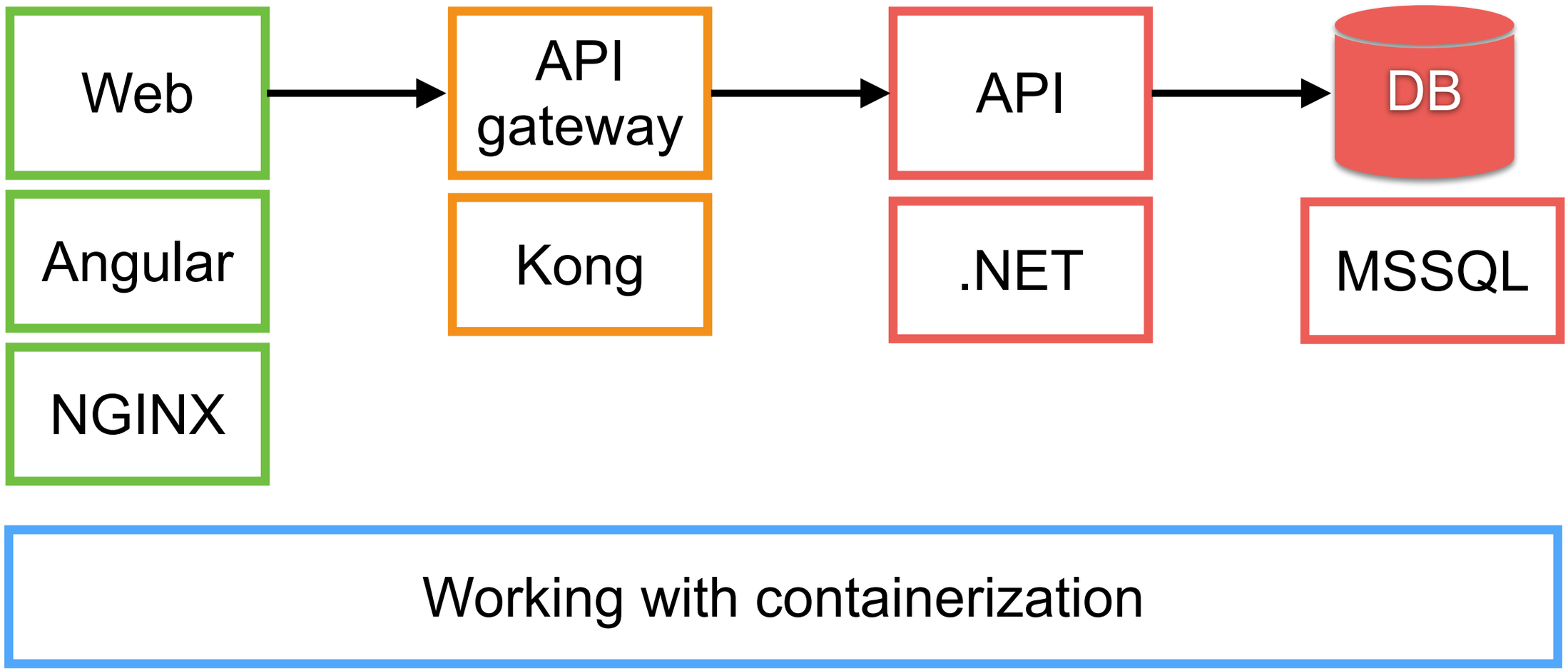
Working with API Gateway



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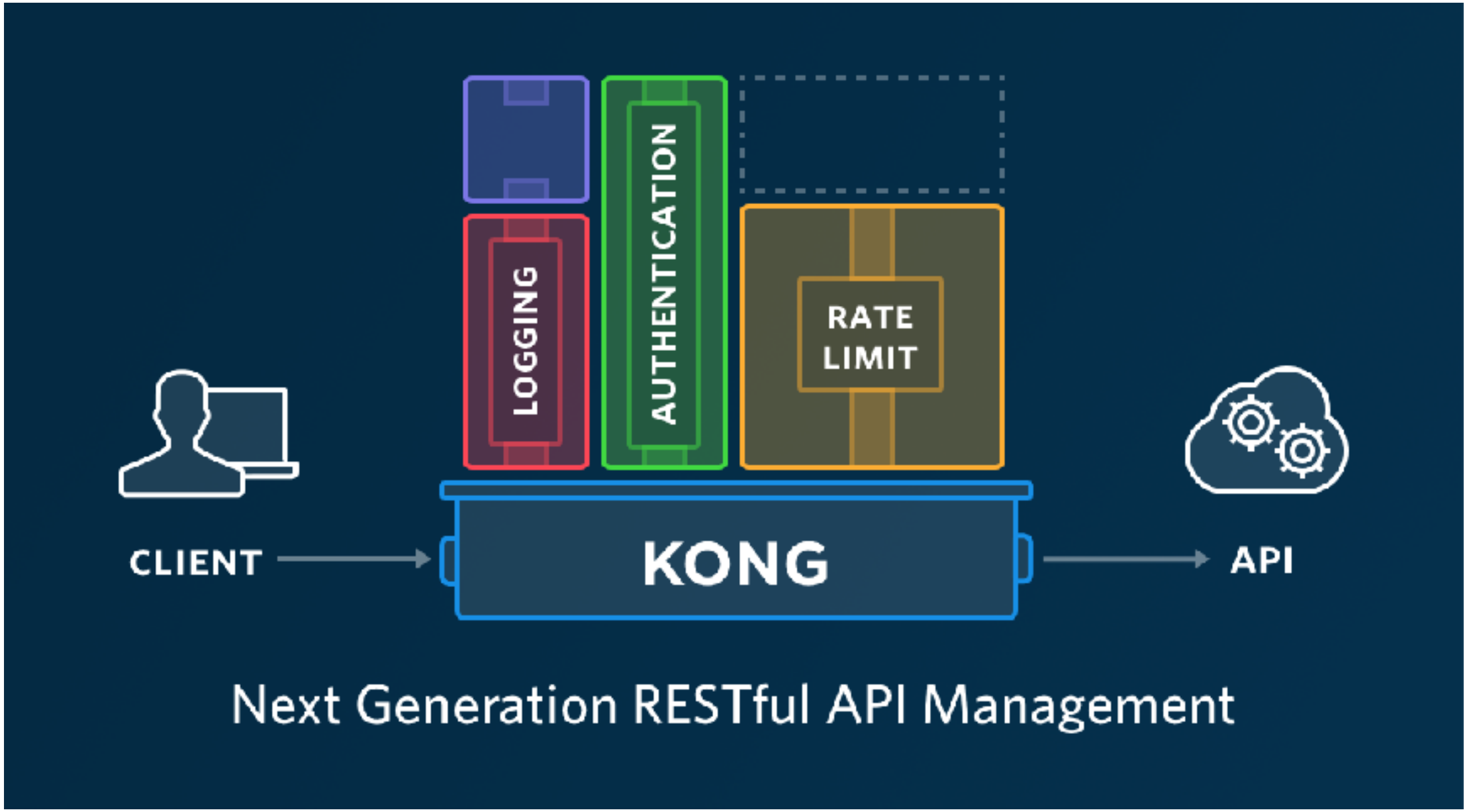

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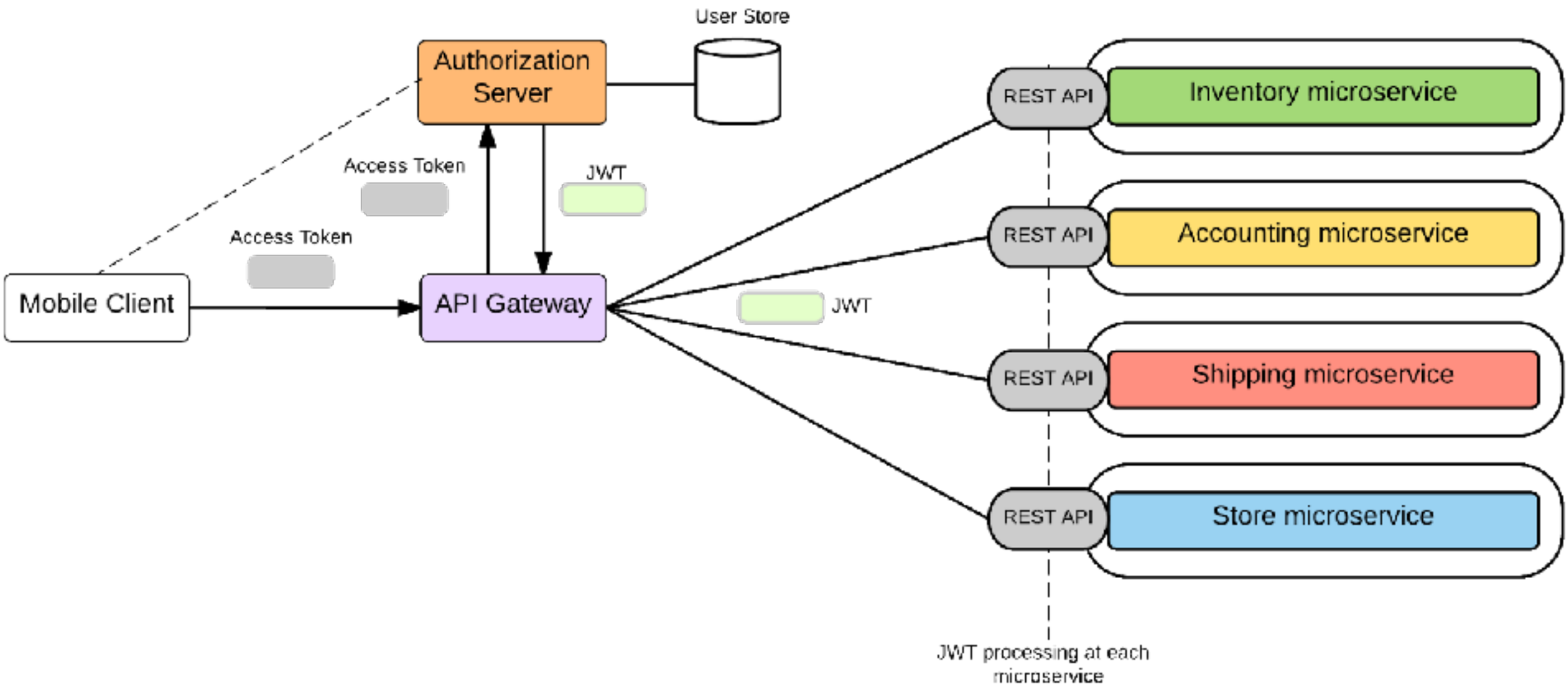

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Q/A

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